

Project Proposal

I. Data summary

- The data set we looked at collects data describing listings activities of homestays from Airbnb in Seattle, WA.
- The data set is cleaned and adjusted to include only categorical and quantitative variables that we believe would be significant in predicting the price of listings from Airbnb in Seattle, and are appropriate for R-studio analysis, excluding data such as property's id, address, url, etc. Categorical variables that contain only 1 level are also excluded from the data.

II. Data analysis

- We chose the following predictors and response variable for the initial model
- The predictor variables that we highlighted are the primary variables we will be using to model with. That is, we will be utilizing priors for these variables and focusing on creating a linear regression model with just the four highlighted variables. The other predictor variables might be used later in our analysis with default priors.

1. Predictors

Quantitative variables:

host_response_rate: the rate of host's responses

host_total_listings_count: The total number of listings managed by the host across all Airbnb platforms

accommodates: number of people the property can accommodate for sleeping arrangements

bathrooms: the total number of bathrooms in the property

bedrooms: number of bedrooms in the property

beds: number of beds in the property

extra_people: the additional charge per person for guests beyond the number of guests included in the base price

minimum_nights: minimum number of nights one has to book

number_of_reviews: number of reviews a listing has received from guests over its lifetime

review_scores_rating: overall rating given to a listing by guests, based on their reviews

reviews_per_month: the average number of reviews a listing receives per month

- *Qualitative variables:*

host_identity_verified: Indicates whether the host's identity has been verified by Airbnb

property_type: the type of property

room_type: the type of room guests will be staying in

bed_type: the type of bed available in the listing

instant_bookable: whether the listing can be booked immediately without the host's approval

cancellation_policy: the host's cancellation policy, which dictates how flexible guests are regarding cancellations

2. **Response variable**

- **price:** price of the property on Airbnb